DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III Topics in Analysis (3TN)

Hilbert Spaces I: The properties of a scalar product

Dr Francois Ebobisse

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0.1 Introduction

The notion of Hilbert spaces, named after the famous German mathematician, David Hilbert (23 January 1862 - 14 February 1943). Hilbert spaces as summarized in Wikipedia, provide a beautiful framework that extends the methods of vector algebra and calculus from the two-dimensional Euclidean plane and three-dimensional space to spaces with any finite and even infinite number of dimensions.

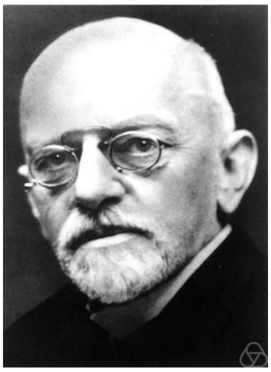

Figure 1: **David Hilbert:** German mathematician, 23 January 1862 - 14 February 1943. His immense contribution in Mathematics involves a broad range of areas.

0.2 Scalar product

Definition 0.1 (Scalar product)
Let H be a vector space over the field $\mathbb{K} = \mathbb{C}$ or $\mathbb{R}$. A scalar product on H is a mapping $\sigma : H \times H \to \mathbb{R}$ with the following properties:

(i) $\sigma(u, u) \geq 0$ for all $u \in H$ with equality holding if, and only if, $u = 0$;

(ii) $\sigma(u, v) = \overline{\sigma(v, u)}$ for all $u,\ v \in H$, where $\overline{\lambda}$ denotes the conjugate of λ;

(iii) $\sigma(\alpha u + \beta v, w) = \alpha\sigma(u, v) + \beta\sigma(v, w)$ for all $u,\ v,\ w \in H$ and all $\alpha,\ \beta \in \mathbb{R}$.

Remark 0.2 It follows easily follows from (i) and (ii) that $v \in (u, v)$ is a sesquilinear form, i.e.,

$$\sigma(u, \alpha v + \beta w) = \overline{\alpha}\sigma(u, v) + \overline{\beta}\sigma(u, w) \quad \forall u,\ v,\ w \in H,\ \forall \alpha,\ \beta \in \mathbb{C}.$$

Given a scalar product σ on H it is customary to write (u, v) in place of $\sigma(u, v)$ for all $u,\ v \in H$ and (u, v) is called the scalar product of u and v.

Example 0.3
(i) In $\mathbb{R}^3$ we have the usual scalar product defined by

$$(\boldsymbol{x}, \boldsymbol{y}) := x_1 y_1 + x_2 y_2 + x_3 y_3 \qquad \forall \boldsymbol{x},\, \boldsymbol{y} \in \mathbb{R}^3.$$

(ii) In the space $M^{3\times 3}$ of square matrices of order 3 we have the scalar product defined by

$$(\boldsymbol{a}, \boldsymbol{b}) := a_{11} b_{11} + a_{22} b_{22} + a_{33} b_{33} \qquad \forall \boldsymbol{a} = (a_{ij}),\, \boldsymbol{b} = (b_{ij}) \in M^{3\times 3}.$$

Definition 0.4 A vector space H equipped with a scalar product $(\cdot, \cdot)$ is called a *scalar product (or pre-Hilbert) space* and is denoted by $(H, (\cdot, \cdot))$.

The following property is fundamental to show that any scalar product induces a norm.

Proposition 0.5 (Cauchy-Schwarz's inequality)
let $(H, (\cdot, \cdot))$ be a scalar product space. Then

$$\boxed{|(u, v)| \leq (u, u)^{1/2} (v, v)^{1/2} \qquad \forall u,\, v \in H}. \tag{1}$$

Proof: Let $u, v \in H$. If $u = 0$ then the inequality (1) is trivial because both sides are zero. Now assume that $u \neq 0$, hence $(u, u) > 0$. Let $\lambda \in \mathbb{K}$. We have

$$\begin{aligned} 0 &\leq (\lambda u + v, \lambda u + v) = |\lambda|^2 (u, u) + \lambda (u, v) + \overline{\lambda} (v, u) + (v, v) = \\ &= |\lambda|^2 (u, u) + \lambda (u, v) + \overline{\lambda (u, v)} + (v, v) = \\ &= |\lambda|^2 (u, u) + 2 Re(\lambda (u, v)) + (v, v), \end{aligned}$$

and by choosing $\boxed{\lambda = -\overline{(u, v)}/(u, u)}$ we finally get $0 \leq -\dfrac{|(u, v)|^2}{(u, u)} + (v, v)$ which implies $|(u, v)| \leq (u, u)^{1/2} (v, v)^{1/2}$. $\square$

Corollary 0.6 *Let $(H, (\cdot, \cdot))$ be a scalar product space. Then the mapping $\|\cdot\| : V \to \mathbb{R}$ defined by*

$$\boxed{\|u\| := \sqrt{(u, u)} \quad \forall u \in V}$$

defines a norm on H, called norm induced by the scalar product

Proof: The first two properties of the norms are trivial. Let us verify property $N3$. From Cauchy-Schwarz's inequality we have

$$\begin{aligned} \|u + v\|^2 &= (u + v, u + v) = (u, u) + 2 Re(u, v) + (v, v) = \\ &= \|u\|^2 + 2 Re(u, v) + \|v\|^2 \leq \|u\|^2 + 2 \|u\| \|v\| + \|v\|^2 = \\ &= (\|u\| + \|v\|)^2. \quad \square \end{aligned}$$

Exercise 0.7 Prove that the scalar product is continuous with respect to the induced norm.

Definition 0.8 A pre-Hilbert space $(H, (\cdot, \cdot))$ is a Hilbert space if H is a Banach space with respect to the norm induced by the scalar product $(\cdot, \cdot)$.

Definition 0.9 Let $(H, (\cdot, \cdot))$ be a pre-Hilbert space.

(i) Two vectors $u, v \in H$ are said to be orthogonal (written $u \perp v$) if $(u, v) = 0$.
A vector $u \in H$ is said to orthogonal to the set $S \subset H$ if $(u, v) = 0$ for all $v \in S$.

(ii) We call *orthogonal complement* of a set $S \subset H$ the set

$$\boxed{S^\perp := \{u \in H : (u, v) = 0 \ \forall v \in S\}}.$$

(iii) Let I be a non-empty set. A family $\{e_i\}_{i \in I}$ of elements in a Hilbert space $(H, (\cdot, \cdot))$ is called *orthonormal system* if

$$\boxed{(e_i, e_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j \end{cases} \qquad \forall i, j \in I}.$$

Example 0.10

(i) Let $u \in H$ with $(u, u) = 1$. The set $\{u\}$ is an orthonormal system in H.

(ii) Let $\delta_{nm} = 1$ if $n = m$ and $\delta_{nm} = 0$ if $n \neq m$ be the Kronecker delta. The family $\{e_n\}_{n=1}^\infty$ with $e_n := \{\delta_{nm}\}_{m=1,2,\ldots}$, is an orthonormal system in the space $\ell^2(\mathbb{C})$.

Proposition 0.11 (Pythagora's theorem)
Let $(H, (\cdot, \cdot))$ be a scalar product space and let $u, v \in H$ with $u \perp v$. Then

$$\boxed{\|u + \lambda v\|^2 = \|u\|^2 + |\lambda|^2 \|v\|^2 \qquad \forall \lambda \in \mathbb{K}}.$$

0.3 Norm induced by scalar product

We have seen in Corollary 0.6 that a scalar product always induces a norm. The following theorem known in the literature as a theorem by **Jordan-Fréchet-von Neumann**, gives a necessary and sufficient condition under which a given norm is induced by a scalar product. That conditions is called **the parallelogram identity**.

Theorem 0.12 (Parallelogram identity)
Let $(V, \|\cdot\|)$ be a normed space. The norm $\|\cdot\|$ is hilbertian, or induced by a scalar product $(\cdot, \cdot)$ if and only if the following identity called parallelogram identity *holds:*

$$\boxed{\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2 \qquad \forall u, v \in V}. \tag{2}$$

In this case, the scalar product is defined by

$$\boxed{(u, v)_{\mathbb{R}} := \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2) \quad \text{if } V \text{ is a real vector space}} \tag{3}$$

and by

$$\boxed{(u, v)_{\mathbb{C}} := \frac{1}{4}\left[(\|u + v\|^2 - \|u - v\|^2) + i(\|u + iv\|^2 - \|u - iv\|^2)\right] \text{ if } V \text{ is a complex vector space}}.$$

Remark 0.13 Notice that the parallelogram identity (2) simply states that the sum of the squares of the length of the four sides of a parallelogram equals the sum of the squares of the length of the two diagonals.

We omit the proof the theorem 0.12 in these lectures!

Exercise 0.14 Let $(H,(\cdot,\cdot))$ be a scalar product space. Let $\{u_n\}$, $\{v_n\} \subset H$ be two sequences such that $\|u_n\| \le 1$, $\|v_n\| \le 1$ for every n and $(u_n, v_n) \to 1$. Prove that $\|u_n - v_n\| \to 0$.

Exercise 0.15 Let $(H,(\cdot,\cdot))$ be an inner product space. Prove that:
(i) If $v \neq 0$, then $\|u+v\| = \|u\| + \|v\|$ if and only if there exists $\lambda > 0$ such that $u = \lambda v$;
(ii) $u \perp v$ if and only if $\|u + \lambda v\| = \|u - \lambda v\|$ for any $\lambda \in \mathbb{K}$, and also if and only if $\|u + \lambda v\| \ge \|u\|$ for any $\lambda \in \mathbb{K}$.

Exercise 0.16 Let $C([-1,1],\mathbb{C})$ be the space of continuous functions from $[-1,1]$ to $\mathbb{C}$. We consider the mapping $\sigma : C([-1,1],\mathbb{C}) \times C([-1,1],\mathbb{C}) \to \mathbb{C}$ defined by

$$\sigma(f,g) := \int_{-1}^{1} f(x)\overline{g(x)}\,dx \qquad \forall f,g \in C([-1,1],\mathbb{C})\,.$$

(a) Show that σ is a scalar product on $C([-1,1],\mathbb{C})$.

(b) Consider the sequence (f_n) defined by $f_n(x) := \begin{cases} -1 & \text{if } -1 \le x \le -\frac{1}{n}\,, \\ nx & \text{if } -\frac{1}{n} \le x \le \frac{1}{n}\,, \\ 1 & \text{if } \frac{1}{n} \le x \le 1\,. \end{cases}$

   (i) Show that the sequence (f_n) is a Cauchy sequence in $(C([-1,1],\mathbb{C}), \|\cdot\|_\sigma)$ where $\|\cdot\|_\sigma$ denotes the norm induced by the scalar product σ.

   (ii) Let $f : [-1,1] \to \mathbb{R}$ be the function defined by $f(t) := \begin{cases} -1 & \text{if } -1 \le x < 0\,, \\ 0 & \text{if } x = 0\,, \\ 1 & \text{if } 0 < x \le 1\,. \end{cases}$

   (iii) Show that the sequence (f_n) converges to the function f in the norm $\|\cdot\|_\sigma$, i.e., $\lim\limits_{n\to\infty} \|f_n - f\|_\sigma = 0$.

   (iv) Deduce that $(C([-1,1],\mathbb{C}),\sigma)$ is not a Hilbert space.

Exercise 0.17 Consider the real vector space $\mathbb{R}^2$ equipped with the norm

$$\|(x,y)\|_\infty := \max(|x|,|y|) \qquad \forall (x,y) \in \mathbb{R}^2\,.$$

(a) Let $C := \{(x,y) \in \mathbb{R}^2 \colon\, x \ge 1\}$. Determine the quantity $\inf\limits_{(x,y)\in C} \|(x,y)\|_\infty$.

(b) Is the infimum in (a) achieved. If that is the case, then find the set of all those vectors $(x,y) \in C$ which achieves the infimum.

(c) Say whether the norm $\|\cdot\|_\infty$ is induced by some scalar product on $\mathbb{R}^2$.

Exercise 0.18 Let $\mathbb{R}_4[X]$ be the vector space of polynomials wih real coefficients and degree less than or equal to 4, equipped with the scalar product defined by

$$(p,q) := \int_{-1}^{1} p(x)q(x)\,dx \qquad \forall p,q \in \mathbb{R}_4[X]\,.$$

Let $S := \{p \in \mathbb{R}_4[X]\colon\ p(-x) = -p(x) \quad \forall x \in [-1,1]\}$.

(a) Find a basis of S.

(b) Find the orthogonal complement $S^{\perp}$ of the subspace S.

In the next lecture, we will study the notion of orthogonal projection onto closed subspaces.